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Review

A review on protein misfolding, aggregation and strategies to prevent related ailments

Tooba Naz Shamsi, Teeba Athar, Romana Parveen, Sadaf Fatima*

Department of Biotechnology, Jamia Millia Islamia, New Delhi 110025, India

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ABSTRACT

This review aims to highlight the fundamental mechanism of protein misfolding leading to protein aggregation and associated diseases. It also aims to anticipate novel therapeutic strategies with which to prevent or treat these highly debilitating conditions linked to these pathologies. The failure of a protein to correctly fold *de novo* or to remain correctly folded can have profound consequences on a living system especially when the cellular quality control processes fail to eliminate the rogue proteins. The core cause of over 20 different human diseases which have now been designated as ‘conformational diseases’ including neurodegenerative diseases such as Alzheimer’s disease (AD), Huntington’s disease (HD) and Parkinson’s disease (PD) etc. A comprehensive study on protein misfolding, aggregation, and the outcomes of the effects of cytotoxic aggregates will lead to understand the aggregation-mediated cell toxicity and serves as a foundation for future research in development of promising therapies and drugs. This review has also shed light on the mechanism of protein misfolding which leads to its aggregation and hence the neurodegeneration. From these considerations, one could also envisage the possibility that protein aggregation may be exploited by nature to perform specific physiological functions in differing biological contexts.

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1. Introduction

The effective functioning of a protein relies upon its well-coordinated interactions at various levels of cellular processes. Biological functions are under surveillance of around 30,000 proteins, a figure typically present in a mammalian cell. Amino acids serve to be the building blocks of a protein to form a linear chain [1]. Although this linear chain is not functional until it undergoes post translational modifications to achieve its native form. Due to amino acids interactions, proteins soon after translation undergoes folding. After synthesis, some proteins fold spontaneously, whereas numerous proteins result in ineffective folding and are disposed to misfolding. Cellular environment is highly crowded thus misfolding intensifies [2,3]. Proper protein folding, trafficking, and degradation of misfolded proteins is under scrutiny of the Protein Quality Control System (PQC) existing in the living organisms and has evolved over the long period of history. Even in the presence of this system many proteins fail to achieve their native state and this results in misfolding. Partial protein unfolding leads to the formation of certain vital intermediates which might rearrange by themselves to build oligomeric aggregates which finally transform to ordered amyloid fibrils [4–8]. Mechanisms like Proteosomal or Lysosomal degradation targets misfolded proteins when molecular chaper-

ones fail to refold them (Fig. 1). Factors influencing the misfolding of proteins are somatic or genetic mutations, ageing, changes in the cellular environment viz. temperature, pH, oxidative stress, and presence of metal ions [9].

The human pathologies linked to the formation of amyloid fibrils, range from Alzheimer disease (AD), in which the aggregates deposit in the brain cells, to type II diabetes where this deposition occurs in the pancreas, to systemic conditions, where aggregates are deposited in multiple organs, including liver, heart etc. In addition, since past two decades, it has been established that it is possible to find these conditions *in-vitro* under which peptides and proteins with no link with disorders can convert into fibrils with all the characteristics of those associated with amyloid pathologies. Improper folding and aggregation of disease specific proteins are among the common mechanisms for the development of neurodegenerative disorders such as Alzheimer’s (AD), Parkinson’s (PD), Huntington’s (HD), Transmissible Spongiform Encephalopathies (TSEs), and Amyotrophic Lateral Sclerosis (ALS) [10–14]. Over the recent times, researchers revealed the presence of a pathogenic mechanism which is commonly related with these disorders. Mechanism being the aggregation of the misfolded protein and their deposition in various parts of the brain leading to amyloidosis of Central Nervous System (CNS) [15]. This results in the slow conversion of highly soluble proteins into insoluble, filamentous polymers with the characteristic presence of cross- β -pleated sheet structures. These structures are then accumulated as amyloid fibrils and later deposits in either cytoplasm or nucleus of brain cells which are affected or in the extracellular space [5]. Besides, to protect against the accumu-

* Corresponding author.

Email addresses: tooba.shamsi@gmail.com (T.N. Shamsi); teebaathar@gmail.com (T. Athar); parveen.romana@gmail.com (R. Parveen); amu.jmi@rediffmail.com, sfatima1@jmi.ac.in (S. Fatima)

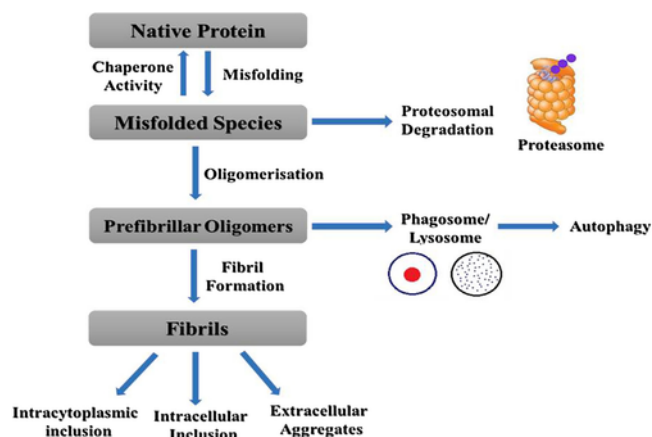


Fig. 1. The illustration of protein misfolding and subsequent aggregation. The native protein gets misfolded and protein quality control system keeps check through Proteosomal and lysosomal degradation of such misfolded proteins. If PQC fails to destruct misfolded proteins it leads to the formation of prefibrillar oligomers and later to fibrils. The accumulation of these fibrils occurs intracytoplasmic, intercellular, and extracellular regions in the body.

lation of misfolded and aggregated proteins cells own a very sophisticated quality control system. Molecular chaperones guarantee that proteins are folded efficiently and keep a check on misfolding and aggregation of non-native proteins [5,16–19].

This review focuses on the mechanisms of protein misfolding, aggregation in consideration of neurodegenerative diseases. The main objective is to provide a facile overview of protein misfolding as a cause of neurodegeneration. Hence, we have focused the most recent reports supporting these ideas and discussed their clinical and biological significance.

2. The intricate mechanism of protein folding

Amino acids are the key to the polymeric sequences of different proteins with typically 300 such blocks. To function, these proteins right after synthesis undergo folding to form compact structures [20]. The sequence of amino acids eventually results in the formation of three-dimensional structure with influence of interactions like Hydrophobic, covalent, and weak forces. Which leads to the formation of an efficiently functioning protein with a native conformation [21,22]. It is now in our knowledge that within the living cells there are various factors supplementary in nature that play role in the folding process like molecular chaperones and folding catalysts [23]. The way a newly synthesized protein which is merely a chain of amino acids converts itself into a folded protein is dependent on the intrinsic properties of amino acids and other contributing influences present in the cellular environment [24]. Research over the years has revealed that folding mechanism is coupled with the process of translation [25,26]. Although, the major folding mechanism of some proteins occurs when proteins are released into cytoplasm from ribosome. Whereas, folding of some other proteins is carried out in endoplasmic reticulum (ER) followed by their translocation through membrane [5]. Folding process is dependent on the environment when folding is carried out. Sometimes, exposed polypeptide regions of a partially folded protein face such an environment where they are vulnerable to meet other unusual cellular molecules. Further investigations suggested that for a short period a non-native interaction is formed to protect the regions which are highly susceptible to aggregation like exposed hydrophobic surfaces [27].

Folding mechanism is also linked with the concept of an “energy landscape” which describes that free energy of a protein influences its conformation. The landscape is often pictured as a funnel because the conformational space of a protein is reduced as it achieves its native form [21,28]. As folding progresses the high chances of disorder of polypeptide chain is reduced significantly, as the more suitable enthalpy is related with stable native-like interactions balance the decreasing entropy as the structure achieves more ordered state. Studies suggests that such a process with efficient folding occurs with smaller proteins, most of them containing less than 100 residues. These small proteins achieve their native conformation without the production of any intermediate states [29]. A classical method is often employed, where a protein can unfold in a solution, such as urea and then placing the protein back to conditions under where native state is achieved back. Complex recovery process of the native structure is often monitored with the help of biophysical techniques like circular dichroism or NMR spectroscopy coupled with stopped or quenched flow methodologies [30].

Other proteins mediate the folding process as well, they are called as Molecular chaperones. They assist the self-assembly process during the catalysis of protein folding. They function by attaching with the partially folded polypeptides and stabilizes them along the way to their correct native state. Without the presence of chaperones, unfolded proteins would be unstable in the cellular environment, frequently folding incorrectly or aggregating into insoluble complexes. Enzymes also facilitates the folding process such as Protein Disulfide Isomerase (PDI) which is involved in the breakage and reformation of disulfide bonds between cysteine residues. Likewise, Peptide prolyl isomerase catalyses isomerisation of peptide bonds involving proline residues [31]. With the help of experimental studies, it is now understandable that how such mechanism can generate a unique fold.

3. Why and how does a protein misfold?

Misfolding occurs when a protein undergoes inefficient folding resulting in formation of a non-functional structure. There can be several reasons behind misfolding like somatic mutations, transcriptional or translational errors, breakdown of folding and chaperone machinery, errors in post-translational modifications, trafficking of proteins, structural alterations due to environmental changes. [1,116,117].

Proteins are under protection during the harsh conditions by many osmolytes which are present in the body [32]. It is now in our knowledge that under optimum conditions, cells possess machinery to avoid protein misfolding. Proteins which unable to achieve their native conformation are assisted by chaperones which help them to fold properly by harvesting energy from ATP [33]. Chaperones can prevent the conformational change from α helix to β sheet and aggregation of these proteins. Thus, chaperones play a major role in the prevention of protein misfolding. Because of environmental factors or overexpression and mutation, native state of a protein is lost and it results in formation of toxic conformation which arises various deadly diseases like Huntington’s disease, Alzheimer’s Disease, prions, cancer, and many neurodegenerative diseases. Misfolding is the phenomenon where folding doesn’t occur efficiently or goes completely wrong [34].

4. Why and how does protein aggregation occur?

Aggregation is produced when hydrophobic amino acid residues or regions of partially folded proteins or intermediates are exposed in the cellular environment that are mainly buried in the native states. Hydrophobic forces essentially are driving forces which result in the generation of unstructured aggregates, oligomers, amyloid fibrils (Fig. 2).

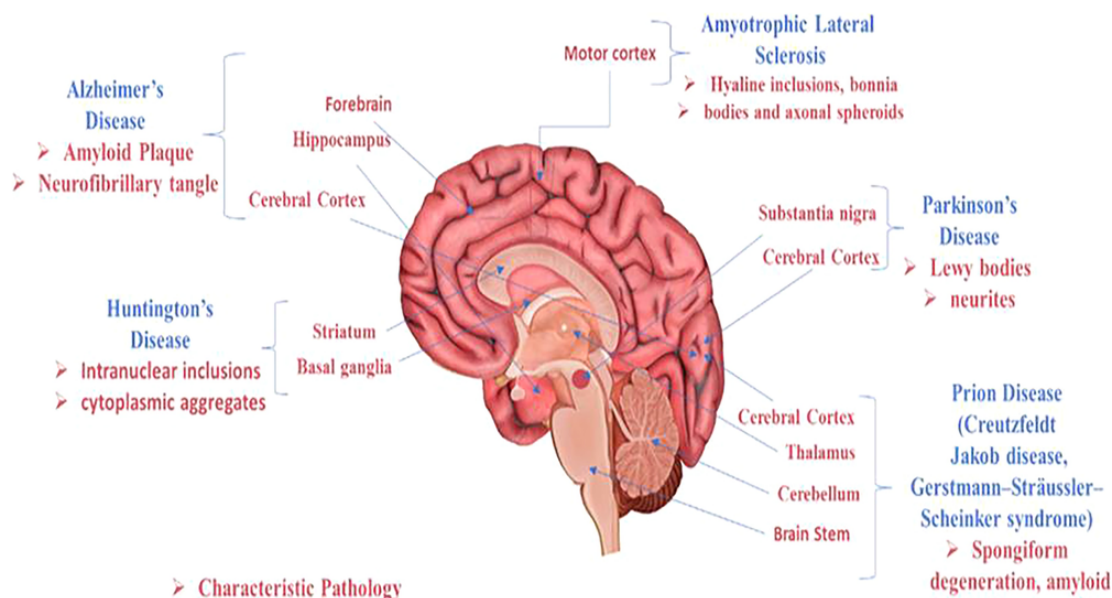


Fig. 2. The above illustration represents the neurodegenerative disorders to their corresponding characteristic pathology and the affected areas of brain. This provides an uncomplicated perspective of the regions affected. The brain diagram was adapted from the web [65].

There are various factors serving as important causal elements that are involved in the mechanism of aggregation namely high hydrophobicity, tendency to form β -sheets, and low net charge [35–39]. Aggregates are made up of fibres which include mechanisms like protein aggregation and formation of inclusion bodies. Now it is well known that protein aggregation and inclusion body formation causes AD, PD, HD, and ALS. Misfolded proteins with a β -sheet conformation (amyloids) constitute the fibres in these aggregates. Degeneracy is linked with misfolded protein deposition. The most possible explanation is that the aggregates characterise an end stage of a cellular cascade, earlier levels in cascade are more likely to be associated with pathogenesis than the inclusions. The understanding of protein aggregation has significantly increased now, and some evidences that are brought forward are helping in logical therapeutics [40,121–123].

5. Amyloid formation as the outcome of protein aggregation

Amyloids are anomalous fibres, proteinaceous in nature that are deposited usually extracellularly in tissues and organs. Structural analysis of these fibres showed that they are chiefly composed of β -sheet with a typical cross- β conformation. Amyloids are highly stable and resilient to degradation. These fibres instead of constituting a structural or supportive role show pathology associated with an array of diseases. Alzheimer's, spongiform encephalopathies, and type II diabetes are renowned examples of amyloid diseases. The interaction of the intrinsically disordered polypeptide islet amyloid polypeptide (IAPP), which is associated with type 2 diabetes (T2D), with the Alzheimer's disease (AD) amyloid- β ($A\beta$) peptide modulates their self-assembly into amyloid fibrils and may link the pathogenesis of these two cell degenerative diseases [41,118–120]. Protofibrils, thin and small filamentous structures primarily composed of β -aggregates are thought to play a role in amyloid formation. It was formerly believed that amyloids are deposited extracellularly in AD and other amyloid disorders. But, in a recently it has been shown that aggregates can form intracellularly also. Proteins including highly helical polypeptides can achieve amyloid conformation under well-defined experimental conditions and hence amyloid formation is expected to be a general property of polypeptides [42].

6. Neurodegeneration as a consequence of protein misfolding and aggregation

Neuronal loss, neuroinflammation, synaptic alterations are some typical features that are displayed by neurodegenerative diseases [10,43,44]. Although, affected regions of the brains vary among diseases (Fig. 1). In AD, hippocampus, amygdala, and cortex regions of brain are primarily affected by neuronal death [44]. In PD, neuronal death occurs in the substantia nigra and dopamine depletion in the striatum [45]. In HD, neuronal loss occurs in the cerebral cortex and striatum [46]. In ALS, the degeneration occurs in lower motor neurons of both spinal cord and brain stem and in upper motor neurons of motor cortex [47]. The pathogenesis in TSEs is characterised by degeneration of spongiform in brain tissue [48]. To understand the association of protein misfolding and aggregation with neurodegeneration, three hypotheses have been proposed, namely (i) the loss-of-function hypothesis (ii) the gain-of-function hypothesis (iii) the brain inflammation hypothesis.

The loss-of-function hypothesis suggests that the normal activity of a protein is lost due to misfolding and aggregation which might lead to neurodegeneration. HD, PD, ALS, and TSE are best examples. In ALS, misfolding and aggregation causes depletion of Cu-Zn superoxide dismutase (SOD1) which results in generation of reactive oxygen species. The gain-of-function hypothesis is most commonly accepted and it says that there is gain of neurotoxic function when a protein undergoes misfolding and aggregation. In vitro observations of neuronal death caused by misfolded proteins and aggregates supports this hypothesis [49–52]. Protein aggregates trigger several mechanisms initiate neurotoxic activity like induction of apoptosis by extracellular aggregates, cell death through disruption and depolarisation of membrane disturbing signal transduction, induction of oxidative stress through production of reactive oxygen species [53–57].

As per the brain inflammation hypothesis, protein aggregates prompt a chronic inflammatory reaction and causes neurodegeneration [58]. The hypothesis was marked by extensive astrogliosis and microglial activation, increase in brain inflammatory proteins level, a de-

crease observed in the occurrence of AD in humans and animal models [59–64].

7. Associated diseases and their characteristic pathology

In this section of review, one of the prominent neurodegenerative diseases are discussed in a brief manner with an insight of their distinctive pathology that renders them as a disorder.

7.1. Alzheimer's disease (AD)

AD is a progressive neurodegenerative disease which is linked with age. Aging is the most important risk factor for neurodegenerative diseases associated with pathological protein aggregation such as Alzheimer's disease. Hence, distinct pathogenesis like accumulation of amyloid-beta-containing plaques in extracellular regions and tau-containing neurofibrillary tangles development intracellular in areas of medial temporal and cortex in afflicted neurons. [66,113].

7.2. Parkinson's disease (PD)

PD is a progressive neurodegenerative disease. The pathogenesis of this second most occurring neurodegenerative disorder is the degeneration or loss of dopaminergic neurons present in the substantia nigra. The neuropathologic feature of PD is the subsequent development of Lewy bodies that are composed of α -synuclein and ubiquitin. Recently, it has been demonstrated that these aggregated proteins could be transferred from one cell to another through extracellular nanovesicles called exosomes. Since beginning, exosomes are considered to be a means of cellular waste removal and hence, actively participating in cell-to-cell communication. To add more, various inflammatory and signaling molecules, as well as small RNAs are selectively packaged in these vesicles. Therefore, aiming these molecules can be a key strategy to prevent PD [114].

7.3. Huntington's disease (HD)

HD is a neurodegenerative autosomal-dominant disorder occurring relatively late in life. It is caused by CAG repeat expansion in IT15 gene resulting in an extended stretch of polyglutamine that falls close to N-terminus of the HD protein huntingtin (htt). The ailment is characterized by psychiatric disturbances, decline in intelligence quotient, uncontrollable motor dysfunction resulting in progressive dementia and eventual death after 15–20 years of disease onset [22].

7.4. Amyotrophic lateral sclerosis (ALS)

ALS is a late-onset fatal neurodegenerative disorder of the human motor system. Degeneration of motor neurons occurs in the motor cortex in the brain, the motor nuclei of brainstem, and in spinal cord anterior horns [67].

7.5. Prion disease

Prion diseases are also known as transmissible spongiform encephalopathies that are characterised by neurodegenerative conditions in humans. The pathogenesis of this disease lies in the conformational alteration of the cellular prion protein denoted as PrP^c into a compulsive form PrP^{TSE}. This transformation is critical to pathogenesis as well as the generation of an infectious prion. Prion diseases are known to affect the central nervous system (CNS) in many species and these diseases are considered to be transmissible [68,69].

8. Potential therapeutics and strategies

Neurodegenerative diseases with its incidence in the population around the world is creating an alarming situation. As discussed above in the review that protein misfolding and aggregation plays a pivotal role in neurodegeneration. Potential therapeutics and strategies against these should be extensively studied [70–72].

8.1. Native state stabilization of proteins

Misfolding and aggregation of proteins could be inhibited by molecules that possess the capability to bind to the native state of a protein. The example of transthyretin, PrP, and Ab explains this strategy best. Transthyretin (TTR), a protein with tetrameric β -strand structure which is predominantly found in the serum and cerebrospinal fluid. Its function is transport of thyroxine (T4) hormone and retinol binding protein [73]. In vitro studies have shown a potential ability of thyroxine to inhibit TTR fibril formation via stabilization of the tetramer [74]. In case of PrPs, it is understood that exploration of compounds is promising that could stabilize PrP^c and confer conformational changes that could block generation of PrP^{TSE} [75,76].

8.2. Blocking the aggregation process

Aggregation of proteins, as we know is one of the chief causes of neurodegeneration. Another approach is to deal with this mechanism through inhibition even after misfolding of proteins occurs. It is understood that in amyloid diseases, some molecules are designed selectively to get attached at the end of growing fibril and cap the growth process. These molecules are designed in a way where their one end binds to fibril while the other end doesn't show any binding interface [1]. Such type of approaches is often labelled to involve β -sheet breakers that has manifested great results in vitro studies [77,78]. These β -blockers can inhibit the polypeptide chain folding in a β -sheet structure. They are short synthetic peptides that comprises self-recognition motif of a misfolded protein. They are known to exhibit the disruption of conformations and aggregation of α -synuclein, islet amyloid polypeptide (IAPP), Ab, PrP, insulin [79–86]. Although, with such promising approach a drawback is also noticed which is that these inhibitors form amyloid fibrils themselves. Through inhibition of intermolecular backbone hydrogen bond formation serves as an alternative approach to block formation of amyloids. Modifications in the backbone like N-methylation offers a steric hindrance in formation of amyloids as well as restricts them to implement a β -strand conformation [87–90].

8.3. Reduction of aggregation-prone species

As universally understood that the rate of intermolecular processes such as aggregation is directly dependent on the concentration. Thus, aggregation process most effectively can be obstructed through reduction of the aggregation-prone species concentration. Also, mutations caused through pathogenesis increases the concentration of aggregation-prone intermediates both unfolded and partially folded. There are two classes which group the compounds that are able to block the aggregation proteins [91–93]. The first class comprises those compounds that blocks monomeric interactions and the other class includes those molecules that heap around the edges of β -sheet aggregates and restricts further growth.

8.4. Depopulation of misfolded proteins

Clearance of misfolded and aggregated proteins is a promising strategy for neurodegeneration diseases, one such strategy includes immunization approach. This strategy was first reported in AD, synthetic Ab aggregates serving as antigens were used to trigger the immune system to produce antibodies to clear them [94]. In transgenic animal models of AD, it has been noticed that immunization strategy shows reduction in amyloid formation, cerebral damage, and behavioural injuries [94–96]. Another approach for depopulation of misfolded proteins entails the use of those antibodies that can recognize conformational epitopes present on oligomeric species [97,98], and subsequently promotes their degradation. It has been observed that antibodies with amyloid recognition capability are present in human sera and they can serve a role in the reduction of amyloid fibrils formation of Ab, immunoglobulin light chain, serum amyloid A protein, IAPP, TTR. Studies have constructed the fact that autoantibodies which can recognize amyloidogenic proteins like Ab, α -synuclein, and PrP are present in humans [99–107].

8.5. Effect of nanoparticle-protein interaction on protein aggregation

Developments in the arena of nanotechnology and medicine have led researchers to study and understand effect of nanoparticles and how their physiochemical properties can render inhibition protein aggregation. Studies are being executed to comprehend the inhibitory effect of NPs on amyloid aggregation as well as its therapeutic outcomes on protein misfolding and its related ailments [21]. The mechanism of amyloid fibrillation is called fibrillogenesis, where the soluble native protein gets misfolded and becomes an insoluble one which contains cross- β -sheets. Nanoparticles with size <30 nm can traverse through the blood-brain barrier. Thus, researches are being manifested to understand NPs' effect on protein aggregation in neurodegenerative diseases involving amyloid-beta peptide ($A\beta$) in Alzheimer's disease [108]. To date, existing drugs in market are unable to cure neurodegenerative diseases. Synthesis of biocompatible and biodegradable NPs can lead to a promising cure and control of amyloid-associated diseases such as AD, PD etc. [109,110].

8.6. Protein disaggregases as prospective therapeutics

Protein disaggregases could serve to be a therapeutic approach because of their property to break down of amyloid structures and soluble forms of toxic oligomers along with recovery and restoration of soluble proteins and their functioning [111,112]. Therefore, protein disaggregases can possibly diminish both toxic gain-of-function and loss-of-function associated with protein misfolding, and subsequently could eliminate species that are likely to be responsible for development of disease [111]. Pairing protein disaggregation with protein degradation could serve to be helpful in elimination of toxic and self-templating species, and synthesis of new proteins that could aggravate any disease promoting loss-of-function [112,115].

9. Conclusions

Conformational alteration of protein is considered the foremost causative link causing aggregation. These changes lead to the exposure of vital amyloidogenic amino acid residues to the surrounding. To conclude, the review aimed to cover protein folding, misfolding, aggregation, associated diseases, and therapies in a comprehensive manner which provides a facile understanding of neurodegenerative diseases.

The mechanism of aggregation demands an in-depth exploration in the future as it is the underlying cause of many neurodegenerative diseases. The article also signifies that a specific structural conformation is important for forming cross- β -sheet rich entities, and that prevention of amyloid aggregation usually results in interruption of hydrophobic interaction which is a predominant stabilising force for amyloids. The incidence in neurodegenerative diseases across the globe is making research in this arena vital as no promising therapy has been found yet for the treatment of such disorders. This study has lead to an idea that such strategies or inhibitors which aim to inhibit the protein self-assembly process can be designed to disrupt the hydrophobic interaction between aromatic or hydrophobic amino acid residues. Hence, the cure to the pathology of such diseases mentioned in this review are directly or indirectly related to protein misfolding and aggregation. Finally, we anticipate the development of novel therapeutic strategies which aim to prevent these highly incurable ailments.

Conflict of interest

The authors have declared that there is no conflict of interest.

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